Jan 2019 Sample Solutions.

Q1 $A = \begin{bmatrix} 2 & -3 \\ 0 & 2 \\ 7 & -2 \end{bmatrix}$ $B = \begin{bmatrix} 1 & -2 & 0 \\ 5 & 1 & 2 \end{bmatrix}$

i) SIZE: ROWS × COLS

$A = 3 \times 2$

$B = 2 \times 3$

ii) $a_{32} = -2$ (row, col)

col 2

$A = \begin{bmatrix} 2 & -3 \\ 0 & 2 \\ 7 & -2 \end{bmatrix}$ row 3

$b_{12} = -2$

$B = \begin{bmatrix} 1 & -2 & 0 \\ 5 & 1 & 2 \end{bmatrix}$ row 1

col 2

iii) B^T ← TRANSPOSE (Rows become cols, cols become rows)

$B^T = \begin{bmatrix} 1 & 5 \\ -2 & 1 \\ 0 & 2 \end{bmatrix}$

iv) $2(A + B^T) = 2\left(\begin{bmatrix} 2 & -3 \\ 0 & 2 \\ 7 & -2 \end{bmatrix} + \begin{bmatrix} 1 & 5 \\ -2 & 1 \\ 0 & 2 \end{bmatrix}\right)$

$= 2\begin{bmatrix} 3 & 2 \\ -2 & 3 \\ 7 & 0 \end{bmatrix} = \begin{bmatrix} 6 & 4 \\ -4 & 6 \\ 14 & 0 \end{bmatrix}$

b) inverse of $\begin{bmatrix} 5 & -2 \\ -7 & 3 \end{bmatrix}$ (a = 5, b = −2, c = −7, d = 3)

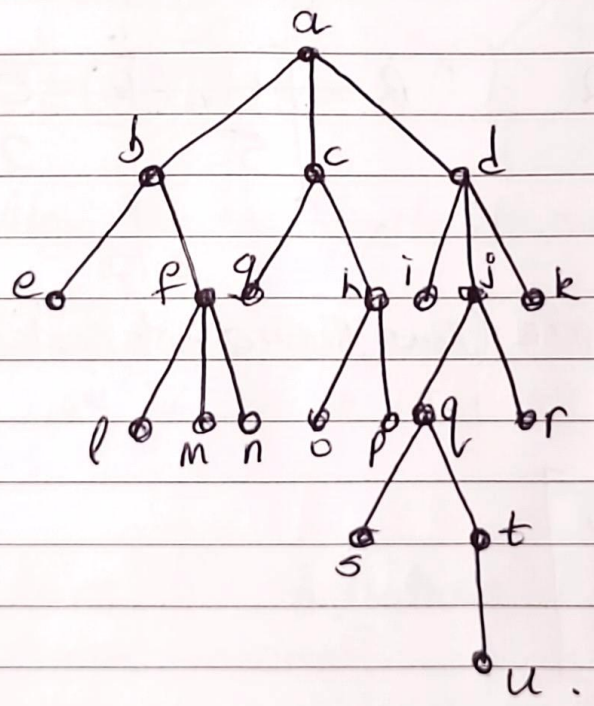
$$A^{-1} = \frac{1}{\det A}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$\det A = ad - bc$$

$$\det A = (5)(3) - (-2)(-7)$$
$$= 15 - 14$$
$$\det A = 1$$

$$A^{-1} = \frac{1}{1}\begin{bmatrix} 3 & 2 \\ 7 & 5 \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ 7 & 5 \end{bmatrix}$$

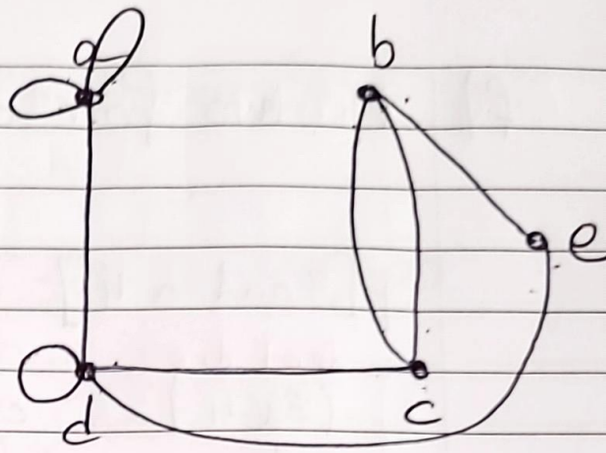
c)



i) Root node = a
ii) children of c = g, h
iii) parent of j = d
iv) number of leaf nodes = 12 (e, l, m, n, g, o, p, i, s, k, r, u)
v) ancestors of q = j, d, a.

d)

	a	b	c	d	e
a	2	0	0	1	0
b	0	0	2	0	1
c	0	2	0	1	0
d	1	0	1	1	1
e	0	1	0	1	0



graph G

degree (A) = 5
" (B) = 3
" (C) = 3
" (D) = 5
" (E) = 2

Note: degree number of edges incident to (touching) a vert
loop = 1 edge, but counts 2 towards the degree.

e) 3, 7, 15, 10, 8, 17, 1, 20

i) Mean: $\bar{x} = \frac{\Sigma x_i}{n} = \frac{3+7+15+10+8+17+1+20}{8}$

$= \frac{81}{8}$

$\bar{x} = \boxed{10.125}$

ii) Standard Deviation: $S^2 = \frac{\Sigma x_i^2 - n(\bar{x})^2}{n-1}$

$S^2 = \frac{3^2+7^2+15^2+10^2+8^2+17^2+1^2+20^2 - 8(10.125)^2}{8-1}$

$= \frac{9+49+225+100+64+289+1+400-820.125}{7}$

$S^2 = 45.27$

$S = \sqrt{45.27} = \boxed{6.7}$

f) i) two fair dice ⇒ total possible number of outcomes $= 6 \times 6 = 36$.

P[3 and a 4]

die1 die2 die1 die2

$(3 \times 4) \quad \text{or} \quad (4 \times 3)$

$\frac{1}{36} + \frac{1}{36} = \frac{2}{36} = \boxed{\frac{1}{18}}$

ii) Build event space for SUM

+	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

$P[\text{SUM} \geq 4] = \frac{33}{36}$

$= \frac{11}{12}$

$P[\text{SUM} \geq 4] = \frac{33}{36}$ or $\frac{11}{12}$

g) $y = 4x^2 - 2x$ use First Principles

$$\frac{dy}{dx} = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}.$$

$f(x) = y = 4x^2 - 2x$

$$\begin{aligned} f(x+h) &= 4(x+h)^2 - 2(x+h) \\ &= 4(x+h)(x+h) - 2x - 2h \\ &= 4(x^2 + 2xh + h) - 2x - 2h \\ &= 4x^2 + 8xh + 4h - 2x - 2h \end{aligned}$$

$$\frac{dy}{dx} = \lim_{h\to 0} \frac{(4x^2+8xh+4h^2-2x-2h)-(4x^2-2x)}{h}$$

$$= \lim_{h\to 0} \frac{4x^2+8xh+4h^2-2x-2h-4x^2+2x}{h}$$

$$= \lim_{h\to 0} \frac{8xh+4h^2-2h}{h}$$

$$= \lim_{h\to 0} 8x+4h-2 \qquad \text{sub in } h=0$$

$$= 8x+4(0)-2$$

$$\frac{dy}{dx} = 8x-2$$

Must show all work.

b) indefinite integrals. $\int x^n . dx = \frac{x^{n+1}}{n+1} + C$

i) $\int (12x^2+6x+3).dx$

$$= \frac{12x^3}{3} + \frac{6x^2}{2} + 3x + C$$

$$= 4x^3 + 3x^2 + 3x + C$$

ii) $\int x^{-2}.dx$

$$\frac{x^{-1}}{-1} + C = -\frac{1}{x} + C$$